S.C.O.U.T. electronics · lab log + Thursday bench plan

I. Rodriguez (ECEN) · logged 2026-10-27 · next bench session Thursday 2026-10-29 · meeting with Dr. Krishnan, Jes and JR on 2026-10-28

Done in the lab, 2026-10-27

☐ ✓ Soldered stacking headers on the **Feather M0 RFM95 (3178)** and the **Adalogger (2922)**; the two boards are now stacked.
☐ ✓ **DS18B20 temperature probe (381):** soldered its 3 bare leads to 3 female jumper wires, twisted each joint, covered with heat-shrink and shrank it with the heat gun. These plug into male–male jumpers on the breadboard, where the 4.7 kΩ pull-up goes.
☐ ✓ **SHT40 leak sensor (4885):** header pins soldered. Ready to connect to the M0.
☐ ✓ **Shore station:** ESP32-S3 Feather (5477) + LoRa FeatherWing (3231) soldered.
☐ ✓ **Dr. Krishnan reviewed the electrical block diagram** (not yet approved).

No second M0 is needed for the shore station. The ESP32-S3 + LoRa FeatherWing *is* the shore station and runs its own firmware (`feather_esp32s3_shore`).

Thursday: what's left, in order

1. **Unpowered checks first** (USB unplugged, multimeter on continuity/Ω):
   - Feather **3V** ↔ **GND**: should **not** beep. A reading near 0 Ω means a solder bridge; find it before powering.
   - Probe: label which female jumper is **red (VDD)**, **yellow (data)**, **black (GND)**. Beep each one end to end through the splice.
   - Look over the header joints for bridges between neighbouring pins.
2. **Feather alone on USB:** meter the **3V** pin to GND → expect **≈ 3.3 V**. Upload a blink test.
3. **Unplug** → wire the temperature probe: red → 3V rail, black → GND rail, yellow → **D12**, 4.7 kΩ between yellow and 3V. Plug in, **touch the probe for 10 s** (warm = unplug at once), read it in the Serial Monitor.
4. **Unplug** → wire the SHT40: VIN → 3V, GND → GND, SDA → **SDA**, SCL → **SCL**. Plug in, read humidity.
5. **Solder the antenna** (see below) → radio test with the shore station.
6. Current measurements, using whatever Dr. Krishnan recommends.

Rules that prevent damage

- Sensors only ever go to the Feather's **3V** pin and **GND**. Never **USB** (5 V) or **BAT**. The M0's pins are 3.3 V only.
- **Unplug USB before changing any wire.** Plug back in only after rechecking.
- **Never run the radio without an antenna.** The buoy firmware transmits, so attach the antenna before loading it. Simple sensor test sketches are fine without one.

What could go wrong when you connect them

Mistake	What happens	How you'd notice / fix
Probe red and black swapped	DS18B20 runs backwards, gets hot, can die in seconds to minutes	Probe warm to the touch → unplug at once. This is why the wires get labeled
Sensor on **USB** or **BAT** instead of 3V	5 V on the SHT40 (rated 3.3 V) and 5 V signals into M0 pins can damage both	Avoided by only using the 3V rail
3V shorted to GND	The regulator current-limits, the board gets warm, the laptop may cut the USB port	Caught by the unplugged 3V↔GND check
Data wire in the wrong pin, or no pull-up	Nothing breaks; the reading just fails (e.g. −127 °C for the DS18B20)	Move the wire, check the 4.7 kΩ
Radio on with no antenna	Transmit power reflects back into the RFM95 and can damage it	Antenna on before the buoy firmware

Everything here runs at 3.3 V from a USB port, so nothing can shock you, and the worst case is one cheap part. A spare DS18B20 is ≈ $10.

Antenna (bench)

- Use the **915 MHz spring antenna (4269)**. Its straight leg goes in the **ANT** hole at the radio end of the M0; solder it from underneath.
- Solder only the **ANT** hole, not the ground pads beside it. Keep the antenna straight and away from other metal.
- A straight 78 mm (≈ 3 in) solid wire in ANT also works.

Getting data to your laptop

- **Sensors:** no shore station needed. USB from the M0 to the laptop, then the Serial Monitor at 115200 baud prints every reading.
- **Radio test:** plug the M0 *and* the shore station into the laptop. The shore window prints `OK buoy=… temp_c_centi=…` for each packet. No WiFi needed.

- The uFL connector (1661) is for the outdoor antenna later. Ask Dr. Krishnan before adding it.
- **No SD card or coin cell needed.** The firmware boots with `sd=FAIL` and the clock marked unset, and keeps going.
- The **128 MB microSD (5250)** was in the Adafruit order, so check that box. The CR1220 is in the Mouser box.

Have ready on the laptop before Thursday

☐ VS Code + PlatformIO installed, repo pulled (`main`)
☐ **Data** USB cables, not charge-only: micro-USB for the M0, USB-C for the ESP32-S3
☐ Buoy, bench build (samples and sends every minute): `cd firmware` → `pio run -e feather_m0_bench -t upload` → `pio device monitor`
☐ Shore station: `pio run -e feather_esp32s3_shore -t upload` → `pio device monitor`
☐ Multimeter, 4.7 kΩ resistor, jumper wires, breadboard, labels/tape

Questions for tomorrow's meeting

Dr. Krishnan

1. What should I measure before connecting the sensors the first time, and what values should I expect (3V↔GND resistance, 3V rail voltage)?
2. How should I measure current: meter in series on USB, or a bench supply on BAT? Which ranges for sleep (µA) versus transmit (≈ 120 mA)? Does the lab have a low-current meter or power profiler?
3. Can you look at my solder joints and heat-shrink splices before I power anything?
4. Is it fine to solder the spring antenna to ANT now and add the uFL later? Should only one be connected at a time?
5. Is the 4.7 kΩ pull-up OK on the breadboard end of the probe's long cable, or should it sit closer to the probe?
6. For the D-cell build later: do you agree with leaving off the fuse and plug?
7. For the Stage 2 bench supply: 3.6 V with what current limit?

Jes and JR

1. Reimbursement: Mouser $27.27 each, Adafruit $47.53 each (in the purchase log).
2. Who can help on Thursday with flashing the firmware and reading the shore station output?
3. When will the hung buoy and lid be ready for an electronics fit check (antenna, SHT40 placement)?